
E-Commerce Customer Analytics & Churn Prediction

Data-Driven Decision Making for Customer Retention

Team Members & Project Roles:

- **Ibrahim Mohamed Ahmed:** Project Presentation, Business Value & Data Cleaning
 - **Ahmed Mady:** Business Intelligence & Power BI Visualizations
 - **Mohamed Fayed:** Machine Learning Modeling & Streamlit Deployment
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1. Executive Summary & Business Importance

- **Why Customer Retention Matters:**
 - Acquiring a new customer costs 5x to 25x more than retaining an existing one.
 - Predicting churn allows businesses to issue targeted retention campaigns before losing revenue.
 - **Project Objective:**
 - Clean and prepare 50,000 customer records for modeling.
 - Uncover key business insights using interactive Power BI dashboards.
 - Build and deploy a Machine Learning model for real-time churn prediction.
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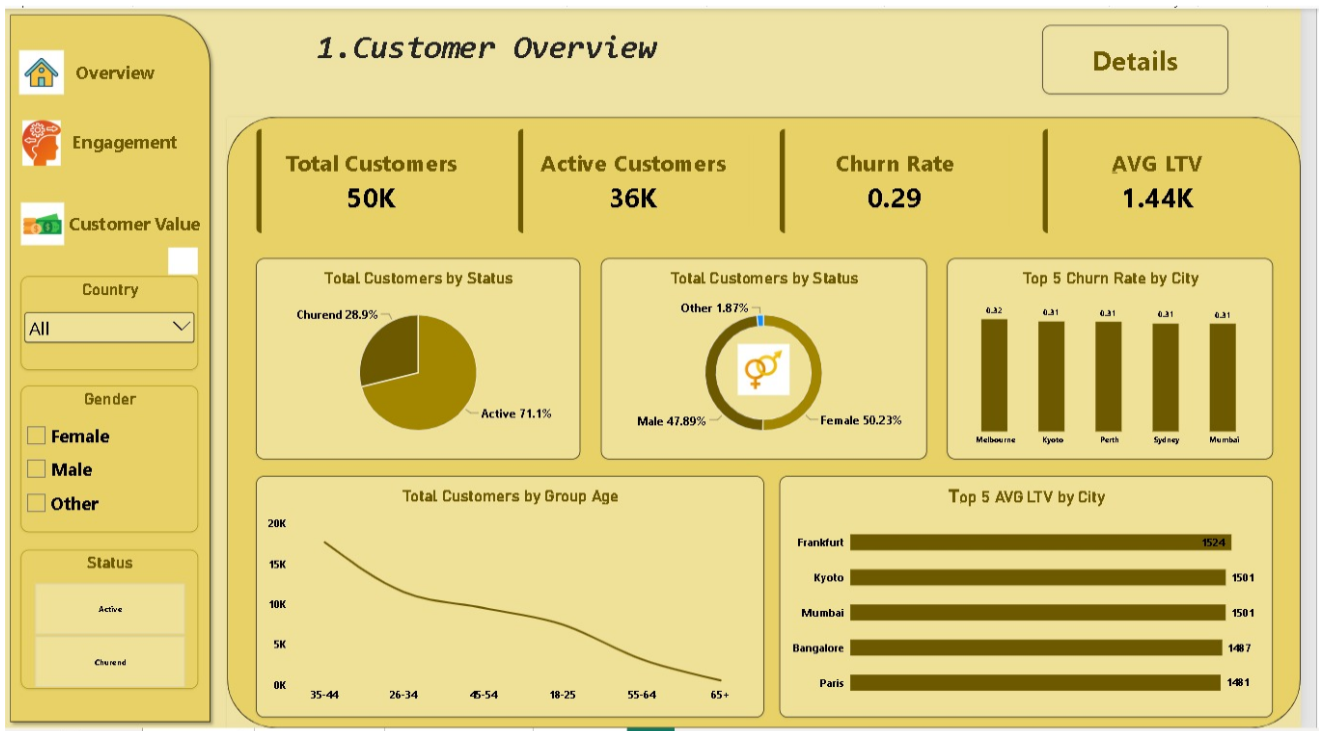
2. Data Preparation & Cleaning (Ibrahim Mohamed)

- **Dataset Overview:**
 - Initial Size: 50,000 Rows | 25 Columns
 - Target Variable: `Churned` (0: Retained, 1: Churned)
- **Cleaning Methodology:**
 - **Duplicates:** Verified and removed duplicate records.
 - **Missing Values:** Median imputation applied to numerical variables; zero-filling for inactive features.
 - **Column Standardization:** Cleaned and formatted all attributes into uniform `snake case`.

- **Final Verification:** 50,000 clean rows with zero null values, keeping essential raw attributes intact.

3. Business Intelligence & Dashboard Overview (Ahmed Mady)

- **Key Performance Indicators (KPIs):**
 - **Total Customers:** 50,000 | **Active:** 36,000 (71.1%)
 - **Churn Rate:** 28.9% | **Average LTV:** \$1,440



- **Key Findings:**
 - Highest customer demographic belongs to the 35–44 age group.
 - Cities with highest Average LTV: Frankfurt (1,524) and Kyoto (1,501).
 - *Live Demo:* Ahmed Mady will now present the interactive Power BI report.

4. Machine Learning Model Comparison (Mohamed Fayed)

- **Evaluated Algorithms:** Logistic Regression vs. Random Forest Classification

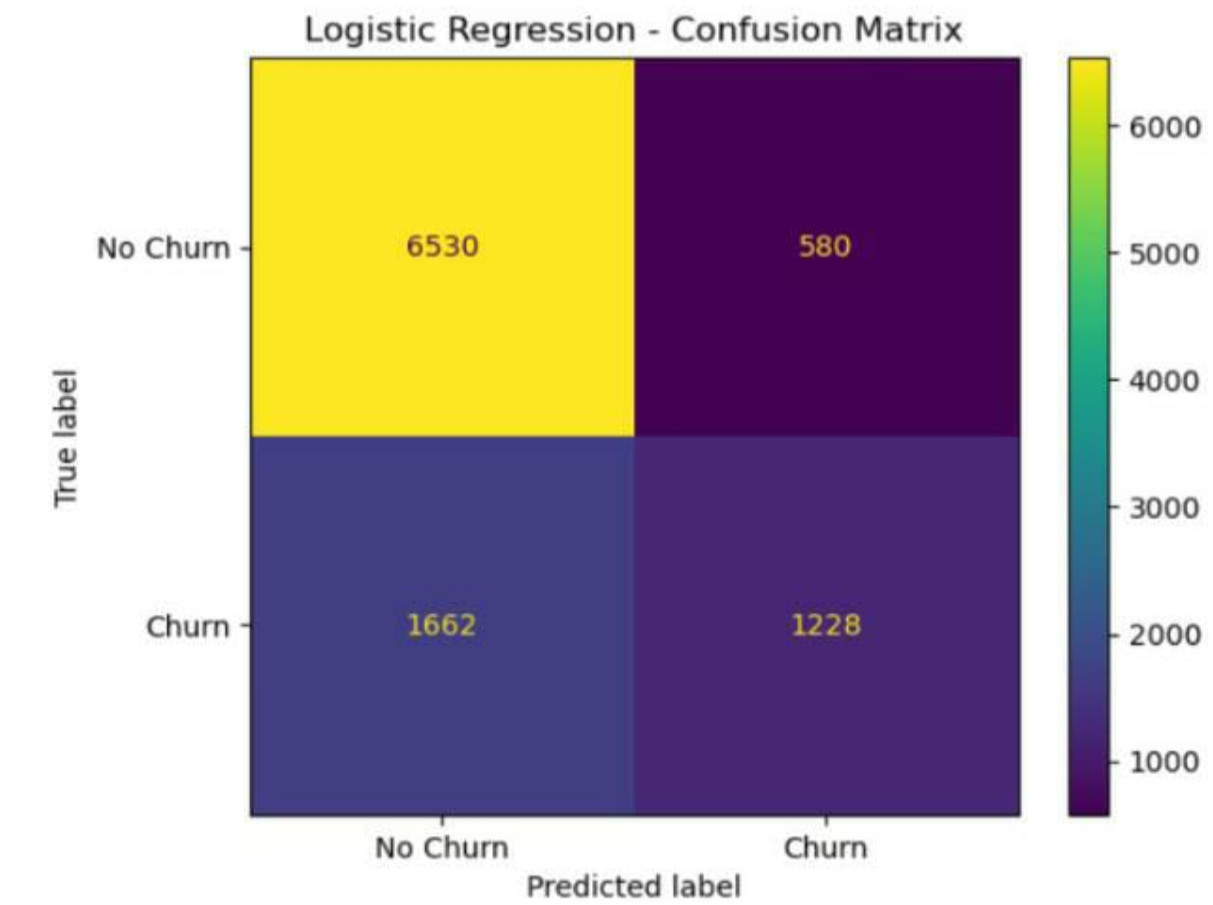
Performance Metric	Logistic Regression	Random Forest (Selected Model)
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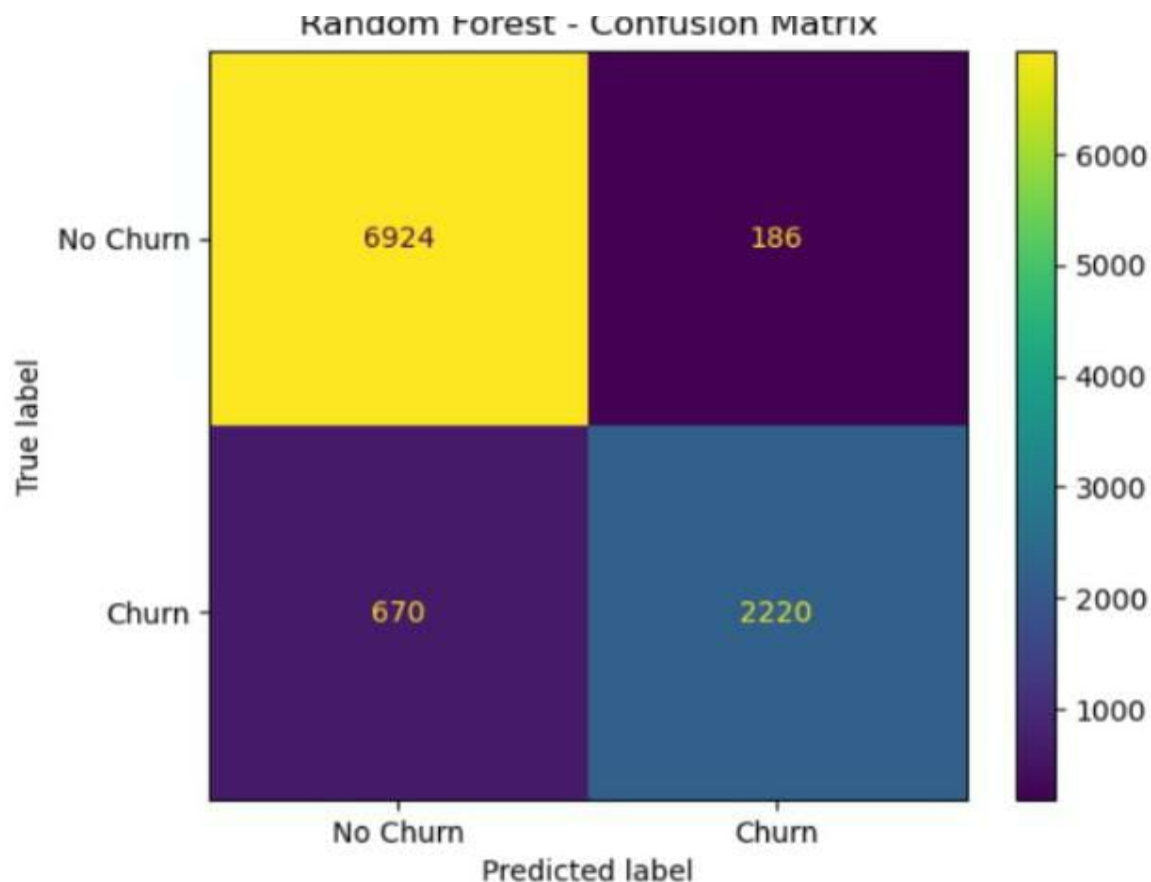
Accuracy	77.58%	91.44%
Precision	67.92%	92.27%
Recall	42.49%	76.81%
F1-Score	52.27%	83.83%

5. Confusion Matrix & Evaluation Analysis (Mohamed Fayed)

- Comparing Model Predictions:

Metric / Outcome	Logistic Regression	Random Forest
True Positives (Correct Churn Detection)	1,228	2,220 (+80.7%)
False Negatives (Missed Churners)	1,662	670 (-59.6%)





6. Model Selection & Business Impact

- **Why Random Forest Was Selected:**
 - **Higher Detection:** Captured 2,220 actual churners compared to only 1,228 by Logistic Regression.
 - **Reduced Risk:** Significantly reduced missed churners (False Negatives), preventing loss of high-value clients.
 - **Cost Efficiency:** Lower False Positives (186 vs 580) ensure retention budgets are not wasted on stable customers.
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7. Streamlit Deployment & Web Application (Mohamed Fayed)

- **App Capabilities:**
 - Accepts real-time customer behavior and support parameters.
 - Outputs exact Churn Probability (e.g., 58.0%) alongside a clear Risk Level indicator.
 - Provides automated, actionable retention recommendations.

8. Strategic Recommendations for Business Growth

1. **Customer Support Protocols:** High service calls (>3) directly signal churn risk; set up priority resolution queues.
 2. **Engagement Rules:** Trigger automated promo codes when Cart Abandonment rates rise.
 3. **High-Value Retention:** Focus retention budget on high-LTV regions like Frankfurt and Kyoto.
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Thank You!